

1) Process Management in Linux

Linux is a multi-user and multi-tasking operating system. This means that many users and many programs can run simultaneously. Each running program is treated as a process and is assigned a unique number called process ID (PID).

process consume system resources such as:

- ① CPU time
- ② main memory (RAM)
- ③ Disk and I/O resource

When unnecessary or fault process continue running they increase system load and reduce performance. To manage this linux provides command like

- ① ps
- ② top
- ③ kill
- ④ killall
- ⑤ pkill

2) Identifying running process

Before killing any process, Admin must first identify which process are running and which

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Name of Process

ones are consuming more system resources.

a) Ps command

The ps (process status) command displays information about active process.

Ex- ps - ef

This command shows:

PID, user who started the process, process-name, CPU usage.

b) top command

The top command displays process in real-time and shows

CPU usage, memory usage, running process. By using ps or top, admin can decide which process should be terminated to reduce system load.

3) Killing process by name

a) Killall command

The killall command is used to terminate all process having the same name.

Syntax = killall process_name

Ex- killall sends a terminates signal to all process with the specified name. It is useful when multiple copies of the same program are running.

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Name of Practical

Ex:- Killall chrome

- 4) Killing a process based on process name
Linux also provides the PKill command to kill process using their name or pattern.

Syntax:- PKill process_name

Theory:- PKill terminates process by matching their names. It is more flexible than Killall because it can match practical names or patterns. It allows users to stop process without knowing their PID.

Ex:- PKill, firefox

- 5) Killing a single process using process PID
Sometimes adam wants to terminate only one specific process without affecting others for this purpose, linux user the kill command along with the process ID.

Step 1 - find the PID

Adam first find the PID of the process using:
PS -ef | grep process_name

Step 2 - Kill the process using PID
Kill PID

ory:- The Kill Command sends a termination signal to the process.
the process is given a chance to close properly.

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Ex:- Kill 2456

- 6) Force Killing a process.
Sometimes a process not respond to normal Kill Commands. It may be stuck or frozen. In such cases, Admin user a force Kill command.

Syntax:- Kill -9 PID

It immediately terminates the process. The process does not get a chance to save data or clean resources. It should be used only when normal Kill fails.

- 7) Signals is Linux process termination.

Linux user signals to control processes.

Important signals are:

- ① SIGTERM (15): Normal termination
- ② SIGKILL (9): Force Termination
- ③ SIGSTOP (: Pause a process
- ④ SIGCONT: Continue a stopped process

The Kill command send these signal to process

- 8) Importance of Killing process

Killing unwanted process is important for:

- ① Reducing cpu load
- ② free memory

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